Ring Theory

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Ring:

Defn: A non-empty set R with two binary operation "addition" $(+)$ & "Multiplication" $(\cdot)$ ie $(R, +, \cdot)$ is said to be a ring. if it ~~follows~~ ~~for~~ satisfy following conditions.

R_1 : $(R, +)$ is an abelian group

R_2 : $(R, \cdot)$ is a semigroup.

R_3 : Set R satisfy Distributive law

ie $\forall\ a, b, c \in R$

(i) $a \cdot (b+c) = (a \cdot b) + (a \cdot c)$ [Left Distribution]

(ii) $(b+c) \cdot a = (b \cdot a) + (c \cdot a)$ [Right Distribution]

For eg: (1) $(\mathbb{Z}, +, \cdot)$ is a Ring where $\mathbb{Z}$ is set of all integers.

(2) $(\mathbb{R}, +, \cdot)$ is a Ring where $\mathbb{R}$ is set of Real numbers

(3) $(\mathbb{Q}, +, \cdot)$ is a Ring. where, $\mathbb{Q}$ is set of Rational numbers

(4) $(\mathbb{C}, +, \cdot)$ is a ring where $\mathbb{C}$ is set of complex numbers.

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Ex: Prove that $R = \{0, 1, 2, 3, 4, 5\}$ is a ring with respect to operation "addition modulo 6" and "multiplication modulo 6"

Soln: Let $R = \{0, 1, 2, 3, 4, 5\}$

and $+_6$ & $\times_6$ are operations.

To show $(R, +_6, \times_6)$ is a ring

we need to show that

R_1 : $(R, +_6)$ is abelian group

R_2 : $(R, \times_6)$ is semigroup

R_3 : Distributive law is holds. by def R

Now By composition table

$+_6$	0	1	2	3	4	5
0	0	1	2	3	4	5
1	1	2	3	4	5	0
2	2	3	4	5	0	1
3	3	4	5	0	1	2
4	4	5	0	1	2	3
5	5	0	1	2	3	4

Table (I)

$\times_6$	0	1	2	3	4	5
0	0	0	0	0	0	0
1	0	1	2	3	4	5
2	0	2	4	0	2	4
3	0	3	0	3	0	3
4	0	4	2	0	4	2
5	0	5	4	3	2	1

Table (II)

To Show : $(R, +_6)$ is abelian group

① **closure law** : from the composition table (I)

since $\forall\ a, b \in R \Rightarrow a +_6 b \in R$

$\therefore$ R satisfy closure property w.r.t to operation

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For eg: $1, 5 \in R \Rightarrow 1 +_6 5 = 0 \in R$

$2, 3 \in R \Rightarrow 2 +_6 3 = 5 \in R$ etc.

(ii) Associative property

From table-(1), it is clear that

$\forall\ a, b, c \in R$, $a +_6 (b +_6 c) = (a +_6 b) +_6 c$

For eg: For 1, 2, 5 $\in R$

$1 +_6 (2 +_6 5) = 1 +_6 (1) = 2$

& $(1 +_6 2) +_6 5 = 3 +_6 5 = 2$

$\therefore 1 +_6 (2 +_6 5) = (1 +_6 2) +_6 5$

$\therefore$ It satisfy Associative property.

(iii) Existance of Identity. ~~property~~

From the table-(1)

Since $0 +_6 0 = 0$, $3 +_6 0 = 3$

$1 +_6 0 = 1$, $4 +_6 0 = 4$

$2 +_6 0 = 2$, $5 +_6 0 = 5$

ie $\forall\ a \in R$, $a +_6 0 = a \Rightarrow$ 0 is identity (additive)

and also $0 \in R$

$\therefore$ There exist an identity $0 \in R$

(iv) Existance of inverse:

We know that $\forall\ a, b \in R$ if $a +_6 b = e = 0$

Then a & b are inverse of eachother.

Here. (from table-(1))

Since $0 +_6 0 = 0$ [0 is inverse of itself]

$1 +_6 5 = 0$ [1 & 5 are inverse of eachother]

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$2 +_6 4 = 0$ [2 & 4 are inverse of each other]

$3 +_6 3 = 0$ [3 is inverse of it self]

and all the inverse are elements of set R

∴ Existance of inverse is hold by set R

(v) Abelian commutative

From the table - (I) [matrix is symmetric]

since $\forall\ a, b \in R \quad a +_6 b = b +_6 a$

∴ It satisfy commutative property.

for ex: For $2, 4 \in R$

$2 +_6 4 = 0 = 4 +_6 2$

∴ $(R, +_6)$ is an Abelian group.

(vi) To show $(R, \times_6)$ is semigroup

(i) closure property

From table - (II)

Since $\forall\ a, b \in R$, $a \times_6 b \in R$

for ex: for $2, 3 \in R \Rightarrow 2 \times_6 3 = 0 \in R$

for $1, 4 \in R \Rightarrow 1 \times_6 4 = 4 \in R$

∴ Closure law is hold by set R w.r.to. operation $\times_6$.

(ii) Associative property:

From table - (II)

Since for $\forall, a, b, c \in R$

$a \times_6 (b \times_6 c) = (a \times_6 b) \times_6 c$

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For eg: let $1, 2, 5 \in R$

$$1 \times_6 (2 \times_6 5) = 1 \times_6 4 = 4$$

& $$(1 \times_6 2) \times_6 5 = 2 \times_6 5 = 4$$

$$\Rightarrow \boxed{1 \times_6 (2 \times_6 5) = (1 \times_6 2) \times_6 5}$$

Here Associative law also hold by set R w.r.t to operation $\times_6$.

$\therefore (R, \times_6)$ is a semigroup.

(III) To show: R satisfy Distributive law

Since for any elements $a, b, c \in R$

$$a *_6 (b +_6 c) = (a \times_6 b) +_6 (a \times_6 c)$$ is hold

by set R $\Rightarrow$ it satisfy distributive law.

For eg: for $2, 3, 5 \in R$

$$2 \times_6 (3 +_6 5) = 2 \times_6 2 = 4$$

& $$(2 \times_6 3) +_6 (2 \times_6 5) = 0 +_6 4 = 4$$

$$\Rightarrow 2 \times_6 (3 +_6 5) = (2 \times_6 3) +_6 (2 \times_6 5)$$ is hold.

Now: since $(R, +_6)$ is abelian group

$(R, \times_6)$ is semigroup

& R satisfy Distributive property

$(R, +_6, \times_6)$ is a ring w.r.t to operation addition modulo 6 & multiplication modulo 6.

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Ex: Show that set of all elements of the type $a+b\sqrt{2}$ where a, b are rational numbers is a ring. w.r.to operation addition & multiplication

(OR) Show that $(R, +, \cdot)$ is a ring.

where $R = \{ a+b\sqrt{2} \mid a, b \in Q \}$

Soln: let R be the set of elements of the type $a+b\sqrt{2}$, where a & b are rational numbers

ie $R = \{ a+b\sqrt{2} \mid a, b \in Q \}$

To show: $(R, +, \cdot)$ is a ring.

we need to show that

(i) $(R, +)$ is abelian group
(ii) $(R, \cdot)$ is semigroup.
(iii) R holds Distributive law.

Now (i) To show $(R, +)$ is abelian group

(i) closure property

let $a, b, c, d \in Q$ [Rational numbers.]

$\therefore$ By defn $a+b\sqrt{2}, \; c+d\sqrt{2} \in R$

Now $(a+b\sqrt{2}) + (c+d\sqrt{2})$

$= (a+c) + (b+d)\sqrt{2}$

$= m + n\sqrt{2}$ [where $m = a+c \in Q$, $n = b+d \in Q$]

$\in R$

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$\therefore$ R satisfy closure property w.r.to addition

(ii) Associative property :

let $x = a+b\sqrt{2} \in R$ where $a, b, c, d, e, f \in Q$

$y = c+d\sqrt{2} \in R$

$z = e+f\sqrt{2} \in R$

Now. $x+(y+z) = (a+b\sqrt{2}) + [(c+d\sqrt{2}) + (e+f\sqrt{2})]$

$= (a+b\sqrt{2}) + [(c+e) + (d+f)\sqrt{2}]$

$= (a+c+e) + (b+d+f)\sqrt{2}$ ——— (1)

& $(x+y)+z = [(a+b\sqrt{2}) + (c+d\sqrt{2})] + [e+f\sqrt{2}]$

$= [(a+c) + (b+d)\sqrt{2}] + [e+f\sqrt{2}]$

$= (a+c+e) + (b+d+f)\sqrt{2}$ ——— (2)

$\therefore$ from (1) & (2)

$\forall x, y, z \in R, \quad x+(y+z) = (x+y)+z$

$\therefore$ R satisfy Associative property w.r.to addition.

(iii) Existance of Identity.

Since $\forall a, b \in Q$ { [Rational numbers] and also $0 \in Q$ }

we know that $a+0 = a$

$b+0 = b$

Now.

for any $x \in R$ such that $x = a+b\sqrt{2}$

Since $x+0 = (a+b\sqrt{2}) + (0+0\sqrt{2})$

$= (a+b\sqrt{2})$

$= x$

$\Rightarrow 0 = (0+0\sqrt{2}) \in R$ is identity of set R

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(iv) Existance of inverse :

We know that for every rational number
i.e. $\forall\ a, b \in \mathbb{Q}$, $a - a = 0$
$b - b = 0$

and since $0 = 0 + 0\sqrt{2}$ is an identity of set R (w.r.t to addition.)

Now, since $0 = 0 + 0\sqrt{2}$

$= (a - a) + (b - b)\sqrt{2}$

$= [a + (-a)] + [b + (-b)]\sqrt{2}$

$$\boxed{0 = (a + b\sqrt{2}) + [(-a) + (-b)\sqrt{2}]} \quad \text{---(3)}$$

Now. Since $a, b \in \mathbb{Q} \Rightarrow -a, -b \in \mathbb{Q}$

$\therefore\ (-a) + (-b)\sqrt{2} \in R$

Thus from above eqn (3)

$\forall\ a + b\sqrt{2} \in R \Rightarrow (-a) + (-b)\sqrt{2} \in R$

is it's inverse.

(v) Commutative property :

let $a + b\sqrt{2},\ c + d\sqrt{2} \in R$ where $a, b, c, d \in \mathbb{Q}$

Now $(a + b\sqrt{2}) + (c + d\sqrt{2})$

$= (a + c) + (b + d)\sqrt{2}$

$= (c + a) + (d + b)\sqrt{2}$

$= (c + d\sqrt{2}) + (a + b\sqrt{2})$

$\Rightarrow$ R satisfy commutative property w.r.to addition

$\therefore$ $(R, +)$ is abelian group.

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(11) To show $(R, \cdot)$ is semigroup.

(1) closure property:

let $m = a+b\sqrt{2} \in R$
$n = c+d\sqrt{2} \in R$ } where $a, b, c, d \in Q$

Now. $m \cdot n = (a+b\sqrt{2}) \cdot (c+d\sqrt{2})$

$= a \cdot c + a \cdot d\sqrt{2} + b \cdot c\sqrt{2} + b \cdot d\, 2$

$= (a+c+2bd) + (a \cdot d + b \cdot c)\sqrt{2}$

Now since $a, b, c, d \in Q$ [Rational numbers]
also $2 \in Q$

$\therefore (a \cdot c + 2bd),\ (ad+bc) \in Q$

$\therefore m \cdot n = (a+b\sqrt{2}) \cdot (c+d\sqrt{2}) \in R$

$\therefore$ R satisfy closure property w.r.to. multiplication

(2) Associative property:

let $x = a+b\sqrt{2} \in R$
$y = c+d\sqrt{2} \in R$
$z = e+f\sqrt{2} \in R$ } where $a, b, c, d, e, f \in Q$

Now. $x \cdot (y \cdot z) = (a+b\sqrt{2}) \cdot [(c+d\sqrt{2}) \cdot (e+f\sqrt{2})]$

$= (a+b\sqrt{2}) \cdot [(c \cdot e + 2d \cdot f) + (c \cdot f + d \cdot e)\sqrt{2}]$

$= ace + 2adf + 2(bcf + bde) + 2(acf + ade)\sqrt{2} + 2(bce + 2bdf)$

Similm $(x \cdot y)z = [(a+b\sqrt{2}) \cdot (c+d\sqrt{2})] \cdot (e+f\sqrt{2})$

$= ace + 2adf + 4bcf + 4bde + 2acf + 2adc + 2bce + 4bef$

$\therefore x \cdot (y \cdot z) = (x \cdot y) \cdot z$ $\Rightarrow$ R is Associative w.r.to multiplication

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$\therefore (R, \cdot)$ is semigroup

(iii) To show : R satisfy Distributive property

let $x = a + b\sqrt{2} \in R$

$y = c + d\sqrt{2} \in R$

$z = e + f\sqrt{2} \in R$

Now. $x \cdot (y+z) = (a+b\sqrt{2}) \cdot [(c+d\sqrt{2}) + (e+f\sqrt{2})]$

$= (a+b\sqrt{2}) \cdot [(c+e) + (d+f)\sqrt{2}]$

$= a((c+e) + (d+f)\sqrt{2}) + b\sqrt{2}((c+e) + (d+f)\sqrt{2})$

$= ac + ae + ad\sqrt{2} + af\sqrt{2} + bc\sqrt{2} + be\sqrt{2} + 2bd + 2bf$ ——— (4)

also. $(x \cdot y) + (x \cdot z) = [(a+b\sqrt{2}) \cdot (c+d\sqrt{2})] + [(a+b\sqrt{2}) \cdot (e+f\sqrt{2})]$

$= (ac + 2bd) + (ad + bc)\sqrt{2} + (ae + 2bf) + (be + af)\sqrt{2}$

$= ac + 2bd + ad\sqrt{2} + bc\sqrt{2} + ae + 2bf + be\sqrt{2} + af\sqrt{2}$ ——— (5)

from (4) & (5)

$x \cdot (y+z) = (x \cdot y) + (x \cdot z)$

$\Rightarrow$ R satisfy Distributive property

$\therefore (R, +, \cdot)$ is a ring. w.r.t addition & multiplication.

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Types of Rings :

① Sub-ring :

A subset S of a ring $(R, +, \cdot)$ is a subring iff S it self is a ring w.r.to same operations

i.e. ① $\forall\ a, b \in S \Rightarrow (a+b) \in S$ & $(a \cdot b) \in S$

② $0 \in S$ [additive identity of S]

③ $\forall\ a \in S \Rightarrow -a \in S$ [additive inverse]

② Commutative Ring :

Let $(R, +, \cdot)$ is a ring under the operation addition and multiplication then it is said to be a commutative ring. If it satisfy commutative property w.r.to multiplication.

i.e. $\forall\ a, b \in R \Rightarrow a \cdot b = b \cdot a$

③ Ring with unity :

A ring $(R, +, \cdot)$ is said to be a ring with unity if there exist an multiplicative identity.

ie ~~If~~ for $a \in R$ $\exists$ some $e \in R$ such that

$$a \cdot e = a = e \cdot a$$

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★ Ring with zero divisors :

A ring $(R, +, \cdot)$ is said to be a ring with zero divisors. if there exist an elements a & b in set R such that

for $a \cdot b = 0 \Rightarrow$ neither $a = 0$ nor $b = 0$

ie. if $a \neq 0, b \neq 0 \in R$ and $a \cdot b = 0$ Then R is ring with zero divisors.

★ Ring without zero divisor :

A ring $(R, +, \cdot)$ is said to be a ring without zero divisor. if there exist ~~so~~ an elements $a, b \in R$ such that $a \cdot b = 0$

$\Rightarrow$ either $a = 0$ OR $b = 0$ or both $a = b = 0$

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Ex: If R is a ring and $a, b \in R$

Then show that

$$(a+b)^2 = a^2 + ab + b\cdot a + b^2$$

$$\neq a^2 + 2ab + b^2$$

Soln: let $(R, +, \cdot)$ is a ring

$\therefore$ for any $a, b \in R \Rightarrow a+b \in R$

[closure property w.r.to addition]

Now.

$$(a+b)^2 = (a+b)\cdot(a+b)$$

$$= a\cdot(a+b) + b(a+b) \quad \text{[Right Distribution]}$$

$$= a\cdot a + a\cdot b + b\cdot a + b\cdot b \quad \text{[left Distribution]}$$

$$\Rightarrow \boxed{(a+b)^2 = a^2 + a\cdot b + b\cdot a + b^2} \quad \text{H.P.}$$

But since R is not commutative ring.

$\therefore a\cdot b \neq b\cdot a \quad \forall, a, b \in R$

$\therefore (a+b)^2 \neq a^2 + a\cdot b + a\cdot b + b^2$

$\Rightarrow (a+b)^2 \neq a^2 + 2ab + b^2$ [need not be true]

Ex:

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Ex: If R is a ring such that $a^2 = a \quad \forall a \in R$ Then show that

(i) $a + a = 0 \quad \forall a \in R$

(ii) $a + b = 0 \Rightarrow a = b \quad \forall a, b \in R$

(iii) R is commutative ring.

Sol: let R is a ring define as

$\forall a \in R, \quad \boxed{a^2 = a}$ ——— (1)

Now (i) To prove that $\forall a \in R, \; a + a = 0$

let $a \in R \Rightarrow a + a \in R$ [closure property w.r.to addition]

$\therefore$ By def^n (1) for $(a+a) \in R$

$\Rightarrow (a+a)^2 = (a+a)$

$\Rightarrow (a+a)\cdot(a+a) = (a+a)$

$\Rightarrow a\cdot a + a\cdot a + a\cdot a + a\cdot a = (a+a) + 0$

($\because 0 \in R$ is additive identity)

$\Rightarrow a^2 + a^2 + a^2 + a^2 = (a+a) + 0$

$\Rightarrow a + a + a + a = (a+a) + 0$ ($\because$ By def^n (1))

$\Rightarrow (a+a) + (a+a) = (a+a) + 0$

$\Rightarrow (a+a) = 0$

$\therefore \forall a \in R \Rightarrow \boxed{a + a = 0}$ ——— (2) H.P.

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(ii) To prove: $\forall\ a, b \in R$, $a + b = 0 \Rightarrow a = b$

let $a, b \in R \Rightarrow a + b \in R$ [closure law w.r.t.o addition]

Now, let $a + b = 0 \Rightarrow a + b = (a + a)$

$\Rightarrow a + b = a + a$ (∵ from previous Result (2))

$\Rightarrow b = a$ [By cancellation law]

∴ $\forall\ a, b \in R$, if $\boxed{a + b = 0 \Rightarrow a = b}$ ——— (3)

(iii) To show R is commutative ring

we need to show that $\forall\ a, b \in R$, $a \cdot b = b \cdot a$

let $a, b \in R \Rightarrow a + b \in R$ [closure law]

$\Rightarrow (a+b)^2 = (a+b)$ (By defn (1))

$\Rightarrow (a+b)(a+b) = a+b)$

$\Rightarrow a^2 + ab + b \cdot a + b^2 = a + b$

$\Rightarrow a + a \cdot b + b \cdot a + b = a + b$ (∵ By defn (1))

$\Rightarrow a + (a \cdot b + b \cdot a) + b = a + 0 + b$

[∵ $0 \in R$ is additive identity]

$\Rightarrow \boxed{a \cdot b + b \cdot a = 0}$ ——— (4)

[∵ By cancellation law]

Since $a, b \in R \Rightarrow (a \cdot b)$ & $(b \cdot a) \in R$ [closure law w.r.t.o multiplication]

also $(a \cdot b) + (b \cdot a) \in R$

Now, since for $ab, ba \in R$, $ab + ba = 0 \Rightarrow a \cdot b = b \cdot a$ (using from (3))

$\Rightarrow$ R is commutative.

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Ex: Prove that
A ring R is without zero divisor
Iff the cancellation law is holds in R.

Soln: let R be ~~the~~ the ring without zero divisor

ie. $\forall\ a, b \in R$, let $a \cdot b = 0 \Rightarrow$ either $a = 0$ or $b = 0$ or both.

To show cancellation law is holds in R
we need to show that: $a \cdot b = a \cdot c \Rightarrow b = c$ $\forall a \neq 0$

let $a, b, c \in R$ such that $a \neq 0$ & $ab = ac$

$\therefore$ let $a \cdot b = ac \Rightarrow a \cdot b - a \cdot c = 0$

$\Rightarrow a \cdot (b - c) = 0$

$\Rightarrow (b - c) = 0 \quad (\because a \neq 0)$

$\Rightarrow b = c$

$\therefore \boxed{a \cdot b = a \cdot c \Rightarrow b = c}$

Hence cancellation law is holds in R

Converse: let R is the ring and consider that R satisfy the cancellation law.

ie. For $a, b, c \in R$, $a \cdot b = a \cdot c \Rightarrow a = b$ ——— (1)

Now, we need to show that R is ring without zero divisor

ie. For $a, b \in R$, if $a \cdot b = 0 \Rightarrow$ either $a = 0$ or $b = 0$ or both

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Let us assume that

R is ring with zero divisor

ie if $a \cdot b = 0 \Rightarrow$ neither $a=0$ nor $b=0$

ie $a \neq 0$ & $b \neq 0$

Now $a \cdot b = 0$

$\Rightarrow a \cdot b = a \cdot 0$ [$\because a \neq 0$, $0 = a \cdot 0$]

$\Rightarrow b = 0$ [By cancellation law]

(but this is not possible as we assume R is ring with zero divisor and $b \neq 0$)

it indicate that our assumption is wrong

$\therefore$ R must be a ring without zero divisor if it satisfy the cancellation law.

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✱ Integral Domain :

Let $(R, +, \cdot)$ is ring w.r.t to addition & multiplication.

Then R, is said to be an integral domain if it satisfy followings

(i) R is commutative ring

ie $\forall\, a, b \in R$, $a \cdot b = b \cdot a$

(ii) R is ring with unity :

ie. if $e \in R$ & $a \cdot e = a$ $\forall\, a \in R$

∴ multiplicative identity exist in R.

(iii) R is ring without zero divisor

ie for $a, b \in R$, $a \cdot b = 0 \Rightarrow$ either $a = 0$ OR $b = 0$ OR both

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Ex: Let $(R, +_5, \times_5)$ is a ring w.r.to operation addition modulo 5 & multiplication modulo 5. where $R = \{0, 1, 2, 3, 4\}$

Then show that R is a finite integral dom

Soln: Let $R = \{0, 1, 2, 3, 4\}$ is a finite set

and $(R, +_5, \times_5)$ is a ring. w.r.to addition modulo 5 & multiplication modulo 5.

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Composition table w.r.to $+_5$ & $\times_5$

$+_5$	0	1	2	3	4
0	0	1	2	3	4
1	1	2	3	4	0
2	2	3	4	0	1
3	3	4	0	1	2
4	4	0	1	2	3

$\times_5$	0	1	2	3	4
0	0	0	0	0	0
1	0	1	2	3	4
2	0	2	4	1	3
3	0	3	1	4	2
4	0	4	3	2	1

① To show: R is commutative ring (w.r.to operation $\times_5$)

Since from the table-(II)

$\forall\ a, b \in R \Rightarrow a \times_5 b = b \times_5 a$

for eg. ⓘ $0 \times_5 3 = 0 = 3 \times_5 0$

ⓘⓘ $3 \times_5 2 = 1 = 2 \times_5 3$

$\therefore$ R is commutative ring.

② To show R is ring with unity

From the table-(II).

Since $0 \times_5 1 = 0$

$1 \times_5 1 = 1$

$2 \times_5 1 = 2$

$3 \times_5 1 = 3$

$4 \times_5 1 = 4$

i.e. for every $a \in R$, since $a \times_5 1 = a$

$\Rightarrow$ 1 is identity w.r.to multiplication

and since $1 \in R$

$\therefore$ multiplicative identity exists in set R

$\therefore$ R is ring with unity.

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(B) To show R is ring without zero divisor

from the table - (II)

∴ Since zero entries are obtained in either column or Row of zero only.

Thus this indicate that

$\forall\ a, b \in R,\quad a \cdot b = 0$

$\Rightarrow$ either $a = 0$
or $b = 0$
or both.

∴ R is ring without zero divisor.

Now. Since $(R, +, \cdot)$ is ring which is Commutative, ring, ring with unity and also ring without zero divisor

$\Rightarrow$ R is an integral domain.

Ex: let $Z = \{0, 1, 2, 3, 4, 5\}$

and $(Z, +_6, \times_6)$ is ring w.r.to operation $+_6$ &

Then show that Z is not an integral domai

Soln: Let $Z = \{0, 1, 2, 3, 4, 5\}$ is a ring with respect to operation addition modulo and multiplication modulo 6.

∴ we form a composition table for Z w.r.to operation $\times_6$ as follows

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$\times_5$	0	1	2	3	4	5
0	0	0	0	0	0	0
1	0	1	2	3	4	5
2	0	2	4	0	2	4
3	0	3	0	3	0	3
4	0	4	2	0	4	2
5	0	5	4	3	2	1

Since from the table

for $2 \neq 0$ & $3 \neq 0$ $\Rightarrow 2 \times_6 3 = 0$

ie. ∃ some element $a, b \in R$ such that

$a \times_6 b = 0$ $\Rightarrow$ neither $a = 0$ nor $b = 0$

ie $a \neq 0, b \neq 0$

$\therefore$ R is not ring without zero divisor

Ring R could not be an integral domain.

w.r.to addition modulo 6 & multiplication modulo 6.

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★ Field :

Let R be the ring w.r.to addition & multiplication operation and let R contain atleast two elements. Then Ring R is said to be a field if it satisfy following conditions.

① R is commutative ring

i.e. for $a, b \in R$, $a \cdot b = b \cdot a$

② R is ring with unity

i.e. $\exists\ e \in R$ s.t. $a \cdot e = a \quad \forall\ a \in R$

(multiplicative identity is exist in R)

③ Multiplicative inverse exist for every non-zero element in R

i.e. $a \neq 0$ & $a \in R$, Then $\exists$ some element $b \in R$ such that $a \cdot b = e$

where e is identity of R w.r.t multiplication operation.

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Ex: Let $(R, +, \cdot)$ is a ring

where $R = \{ a+b\sqrt{2} \mid a, b \in \mathbb{Q} \}$

where $\mathbb{Q}$ is set of rational numbers, then

Show that R is a field w.r.to some operations

Soln: Let $R = \{ a+b\sqrt{2} \mid a, b \in \mathbb{Q} \text{ (rational numbers)} \}$

is a ring w.r.to addition & multiplication

① To show R is commutative ring [w.r.to multiplicat

let $\forall x, y \in R$, where $x = a+b\sqrt{2} \in R$

$y = c+d\sqrt{2} \in R$

and, a, b, c, d $\in \mathbb{Q}$ (Rational numbers)

Now $x \cdot y = (a+b\sqrt{2}) \cdot (c+d\sqrt{2})$

$= ac + 2bd + (ad + bc)\sqrt{2}$

$= c \cdot a + 2d \cdot b + (d \cdot a + c \cdot b)\sqrt{2}$

$= (c+d\sqrt{2}) \cdot (a+b\sqrt{2})$

$= y \cdot x$

$\Rightarrow$ R is commutative ring.

② To show R is Ring with unity:

let $a, b \in \mathbb{Q}$ are rational number

$\Rightarrow a+b\sqrt{2} \in R$

also $1, 0 \in \mathbb{Q}$ are rational numbers

and Hence $1+0\sqrt{2} \in R$

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Now for $a+b\sqrt{2} \in R$

Since $(a+b\sqrt{2}) \cdot 1 = (a+b\sqrt{2})$

$\Rightarrow (a+b\sqrt{2}) \cdot (1+0\sqrt{2}) = (a+b\sqrt{2})$

$\Rightarrow 1+0\sqrt{2} = 1$ is an ~~id~~ multiplicative identity in R

$\therefore$ R is Ring with unity

(iii) To show multiplicative inverse exist in R

Let $a \neq 0$, $b \neq 0$ are two non-zero rational numbers Then $a+b\sqrt{2} \neq 0$ &

and $\boxed{a+b\sqrt{2} \in R}$

Since we know that $\boxed{1+0\sqrt{2} \in R}$ is identity w.r.to. multiplication.

let us consider,

1 $(a+b\sqrt{2}) \cdot (x+b\sqrt{2}) = 1+0\sqrt{2}$

i.e. $(x+b\sqrt{2})$ is multiplicative inverse of element $(a+b\sqrt{2})$

Now we need to show that $(x+b\sqrt{2}) \in R$

Since $(a+b\sqrt{2}) \cdot (x+b\sqrt{2}) = 1+0\sqrt{2}$

$\Rightarrow ax+ay\sqrt{2}+bx\sqrt{2}+2by = 1+0\sqrt{2}$

$\Rightarrow (ax+2by)+(ay+bx)\sqrt{2} = 1+0\sqrt{2}$

equating coefficient of 1 & $\sqrt{2}$ we get

$ax+2by = 1$ ——(1)

& $ay+bx = 0$ ——(2)

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from (2): $ay = -bx$

$\Rightarrow \boxed{y = -\frac{b}{a}x}$ ——— (3)

$\therefore$ (1) $\Rightarrow ax + 2by = 1 \Rightarrow ax + 2b\left(\frac{-b}{a}\right)x = 1$

$\Rightarrow ax - \frac{2b^2}{a}x = 1$

$\Rightarrow x\left(\frac{a^2 - 2b^2}{a}\right) = 1$

$\Rightarrow \boxed{x = \frac{a}{a^2 - 2b^2}}$

$\therefore$ (3) $\Rightarrow y = \frac{-b}{a}(x) = \frac{-b}{a^2 - 2b^2}$

$\Rightarrow$ Since $a, b \in \mathbb{Q} \Rightarrow \frac{a}{a^2-2b^2} = x \in \mathbb{Q}$

& $\frac{-b}{a^2-2b^2} = y \in \mathbb{Q}$

$\therefore x + y\sqrt{2} = \frac{a}{(a-2b^2)} + \frac{-b}{(a^2-2b^2)}\sqrt{2} \in R$

$\therefore \forall\ a + b\sqrt{2} \in R\ \exists$ an inverse $x + b\sqrt{2}$ in R

where $x = \frac{a}{a^2-2b^2}$ & $y = \frac{-b}{a^2-2b^2}$

$\therefore$ multiplicative inverse exist in R

Now since R is commutative ring, R is ring with unit and also multiplicative inverse exist in R

$\Rightarrow R$ is a field.

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Ex: let $M = \left\{ \begin{bmatrix} a & b \\ -b & a \end{bmatrix} / \; a, b \in R \right\}$ is a ring w.r.to operation matrix addition and matrix multiplication, then show that M is a field.

Soln: let $M = \{ A / A = \begin{bmatrix} a & b \\ -b & a \end{bmatrix}$ where $a, b \in R$ is ring w.r.to matrix addition and matrix multiplication.

① To show: M is commutative ring.

let $A = \begin{bmatrix} a & b \\ -b & a \end{bmatrix}$; $B = \begin{bmatrix} \alpha & \beta \\ -\beta & \alpha \end{bmatrix} \in M$

where $a, b, \alpha, \beta \in R$

i.e. $\forall A, B \in R$, $A \cdot B = \begin{bmatrix} a & b \\ -b & a \end{bmatrix} \times \begin{bmatrix} \alpha & \beta \\ -\beta & \alpha \end{bmatrix}$

$$= \begin{bmatrix} a\alpha - b\beta & a\beta + b\alpha \\ -b\alpha - a\beta & -b\beta + a\alpha \end{bmatrix} \quad \text{---} ①$$

also $B \cdot A = \begin{bmatrix} \alpha & \beta \\ -\beta & \alpha \end{bmatrix} \cdot \begin{bmatrix} a & b \\ -b & a \end{bmatrix}$

$$= \begin{bmatrix} \alpha a - \beta b & \alpha b + a\beta \\ -\beta a - \alpha b & -\beta b + \alpha a \end{bmatrix} \quad \text{---} ②$$

from ① & ② $A \cdot B = BA$

$\therefore$ M is commutative ring.

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② To show M is ring with unity

we know that $0, 1, \in R$ (Real Numbers)

$\therefore \begin{bmatrix} 1 & 0 \\ -0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \in M$

Now for any matrix $\begin{bmatrix} a & b \\ -b & a \end{bmatrix} \in M$ where $a, b \in R$

Since $\begin{bmatrix} a & b \\ -b & a \end{bmatrix} \times \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} a & b \\ -b & a \end{bmatrix}$

$\Rightarrow \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ is identity in M

∴ M is ring with unity.

To show M have multiplicative inverse

let us consider $\begin{bmatrix} a & b \\ -b & a \end{bmatrix} \times \begin{bmatrix} x & y \\ -y & x \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

we need to show that matrix $\begin{bmatrix} x & y \\ -y & x \end{bmatrix} \in M$.

let $\begin{bmatrix} x & y \\ -y & x \end{bmatrix}$ inverse of $\begin{bmatrix} a & b \\ -b & a \end{bmatrix}$

$\therefore \begin{bmatrix} a & b \\ -b & a \end{bmatrix} \times \begin{bmatrix} x & y \\ -y & x \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

$\Rightarrow \begin{bmatrix} ax - by & ay + bx \\ -bx - ay & -by + ax \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

$\Rightarrow \begin{bmatrix} (ax - by) & (ay + bx) \\ -(ay + bx) & (ax - by) \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

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$\Rightarrow$ equating corresponding term

$$ax - by = 1 \quad \text{---(1)}$$

$$\& \; ay + bx = 0 \quad \text{---(2)}$$

Solving eqn (1) & (2) simultaneously, we get

$$x = a/a^2+b^2, \quad y = -b/a^2+b^2$$

Since $a, b \in R \Rightarrow x = \frac{a}{a^2+b^2} \in R$

$$y = \frac{-b}{a^2+b^2} \in R$$

$$\therefore \begin{bmatrix} x & y \\ -y & x \end{bmatrix} = \begin{bmatrix} \left(\frac{a}{a^2+b^2}\right) & \left(\frac{-b}{a^2+b^2}\right) \\ -\left(\frac{-b}{a^2+b^2}\right) & \left(\frac{a}{a^2+b^2}\right) \end{bmatrix} \in M$$

$\Rightarrow$ Multiplicative inverse is exist in M.

Now since M Ring $(M, +, \times)$ is is commutative ring, Ring with unity and multiplicative inverse of each non-zero matrices exist in M.

$\Rightarrow$ M is a field.

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Lattice

Partial order Relation:

A relation R on a set A is said to be partial order relation if

1. R is reflexive
2. R is antisymmetric
3. R is transitive

and the set A is called as POSET.

Cover:

Let $(A, \leq)$ is a POSET.

Where "$\leq$" is partial order relation ~~defined as~~ Then an element $b \in A$ is said to be a cover of element $a \in A$ iff $a < b$ and there does not exist any element $x \in A$ such that $a < x < b$

Hasses Diagram:

A partial order relation on a set A can be represented by diagram is known as Hasse diagram. we represent an element of set A by a point or small circle and if b covers a then we take b at higher level that a and join a & b by a straight line so that we get well formed diagram known as Hasses diagram.

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Ex: Draw a Hasses diagram representing power set of set $A = \{a, b, c\}$

Soln: Let $A = \{a, b, c\}$

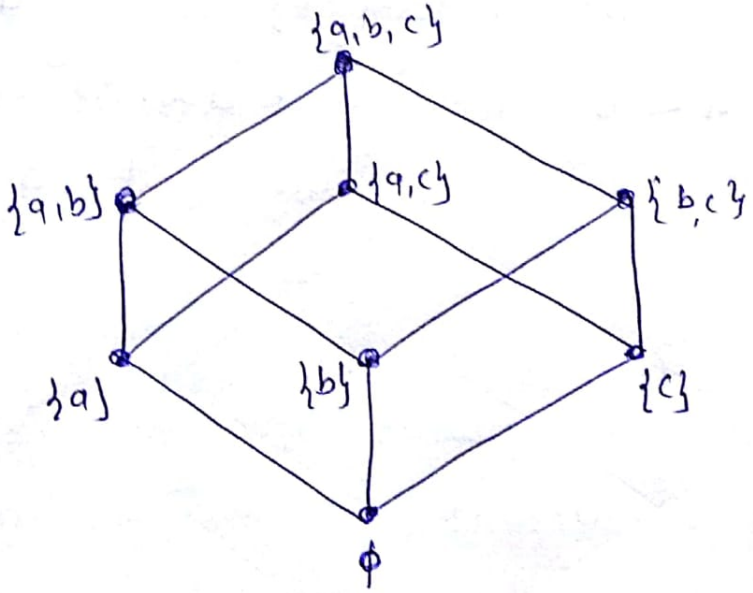
A power set of set A is the set of all subsets of set A.

$\therefore P(A) = \{\phi, \{a\}, \{b\}, \{c\}, \{a,b\}, \{a,c\}, \{b,c\}, \{a,b,c\}\}$

Since The element of set P(A) are related w.r.to Relation "$\subseteq$" (subset)

$\Rightarrow (P(A), \subseteq)$ is partialy order set

and It's Hesses diagram is as follows

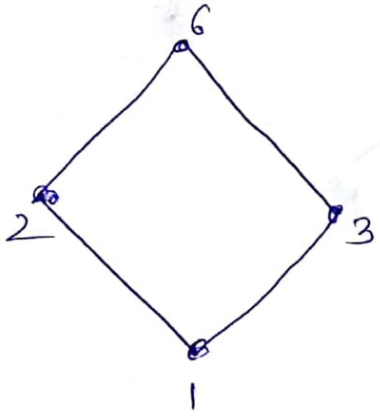


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Ex: For any integer n, let D_n denotes the set of all divisors of n,
Then find Hassed diagram for $n=6$, $n=8$, $n=20$, $n=36$, $n=105$, $n=75$.

Soln: let n is any integer
& D_n denotes the set of divisor of n

① For $n=6 \Rightarrow D_6 = \{1, 2, 3, 6\}$



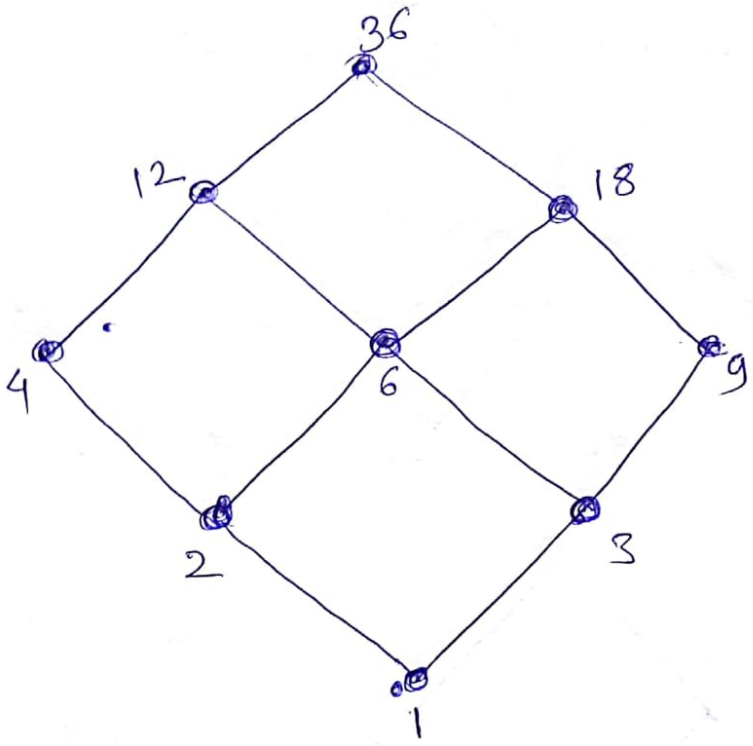
② For $n=8$: $D_8 = \{1, 2, 4, 8\}$



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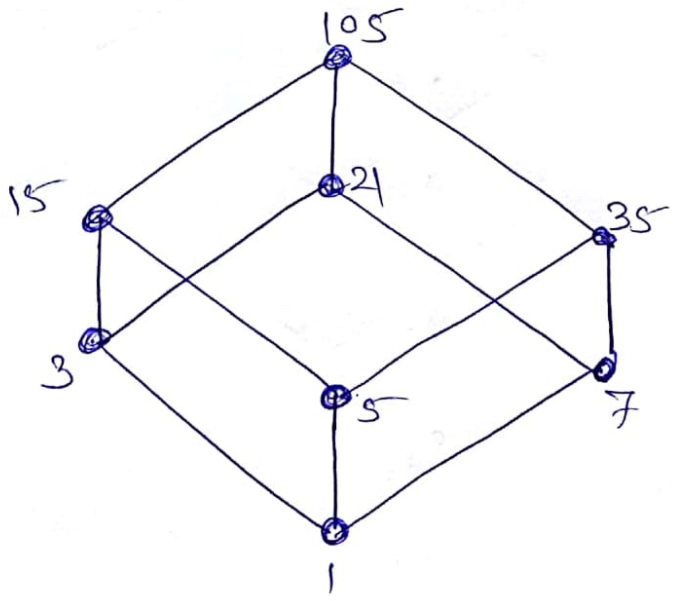
(iii) For $n = 36$:

$D_{36} = \{1, 2, 3, 4, 6, 9, 12, 18, 36\}$



(iv) For: $n = 105$.

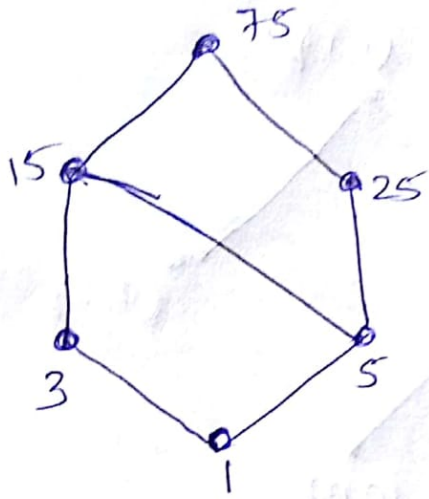
$D_{105} = \{1, 3, 5, 7, 15, 21, 35, 105\}$



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(v) For $n = 75$

$D_{75} = \{1, 3, 5, 15, 25, 75\}$



Ex: Draw a lattice [Hasses diagram] for

$(L, \leq)$, $(L^2, \leq)$ & $(L^3, \leq)$

where $L = \{0, 1\}$

Soln: let $L = \{0, 1\}$

$\therefore (L, \leq) = \{0, 1\}$



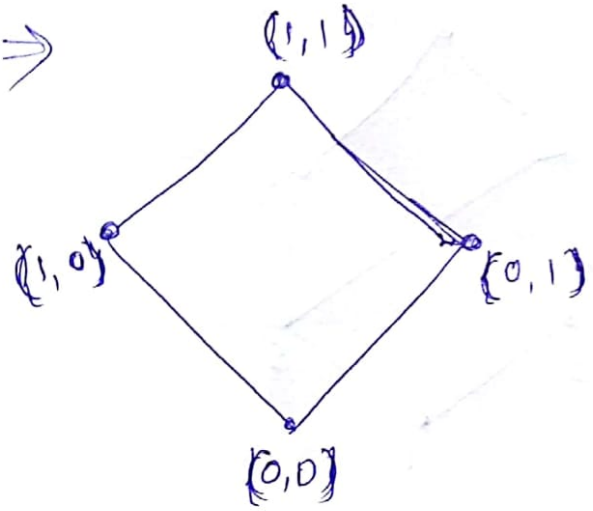
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Now $L^2 = L \times L$

$= \{(0,0), (0,1), (1,0), (1,1)\}$

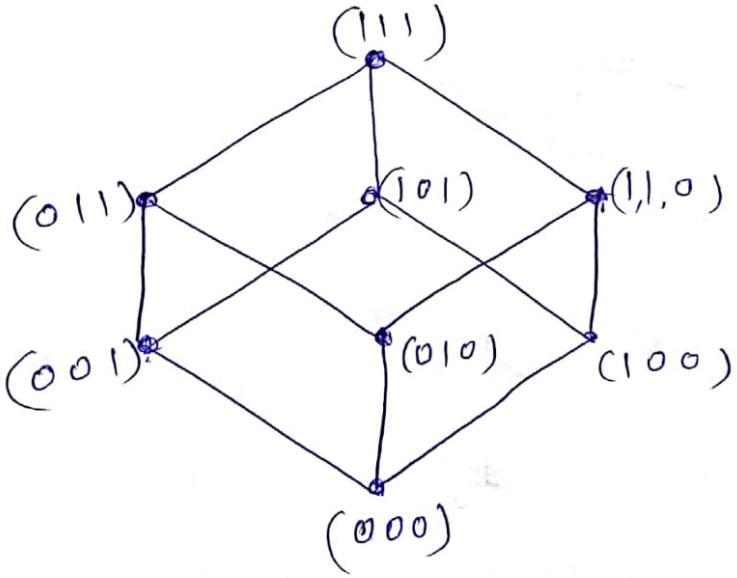
For

$\therefore \{L^2, \leq\} \Rightarrow$



Now $L^3 = L^2 \times L \times L$

$= \{(0,0,0), (0,0,1), (0,1,0), (0,1,1)$
$(1,0,0), (1,0,1), (1,1,0), (1,1,1)\}$



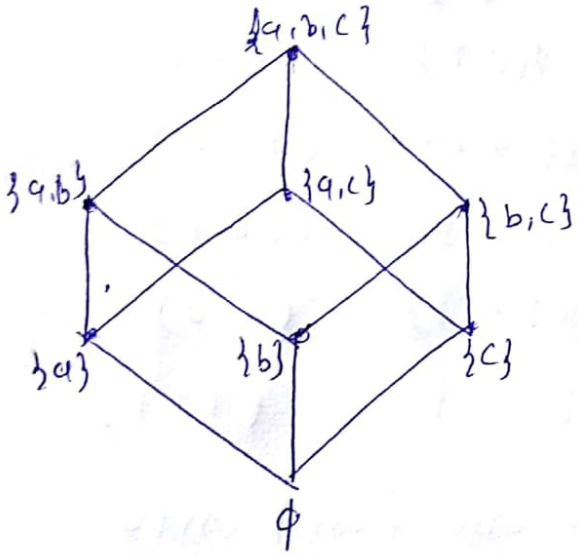
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Ex: let $A = \{a, b, c\}$ and $L = \{0, 1\}$

let $P(A)$ denotes the power set of set A. then show that $(P(A), \subseteq)$ & $(L^3, \leq)$ are isomorphic.

Soln: let $A = \{a, b, c\}$

$\therefore P(A) = \{\phi, \{a\}, \{b\}, \{c\}, \{a,b\}, \{b,c\}, \{a,c\}, \{a,b,c\}\}$

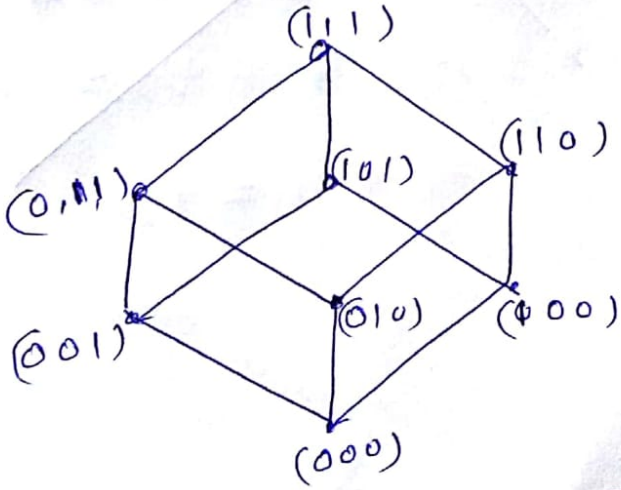


Lattice $(P(A), \subseteq)$

Now: let $L = \{0, 1\}$

$\therefore L^3 = L \times L \times L$

$= \{(000), (001), (010), (011), (100), (101), (110), (111)\}$



Lattice $(L^3, \leq)$

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let f is a function mapping from $P(A)$ to L^3

i.e.

$$f : (P(A), \subseteq) \longrightarrow (L^3, \leq)$$

Since from the lattices of $P(A)$ & L^3 it is clear that

$$f(\phi) = (0\,0\,0)$$
$$f(\{a\}) = (0\,0\,1)$$
$$f(\{b\}) = (0\,1\,0)$$
$$f(\{c\}) = (1\,0\,0)$$
$$f(\{a,b\}) = (0\,1\,1)$$
$$f(\{a,c\}) = (1\,0\,1)$$
$$f(\{b,c\}) = (1\,1\,0)$$
$$\& \quad f(\{a,b,c\}) = (1\,1\,1)$$

$\Rightarrow$ f is one one - and onto

and Hence it is isomorphic

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